

link video

Name	ID
Manar Abdelmonem Reyad	221001939
Soad Auf	221001449

Mini-HPC & Hybrid HPC–Big Data Cluster

Distributed Computing Project

3

VMs

6

MPI Processes

2

ML Tasks

Project Overview

Task 1 — Mini-HPC Cluster

- 3 VirtualBox VMs (Ubuntu 20.04)
- OpenMPI 4.1.6 (compiled from source)
- Python mpi4py — 6 distributed processes
- Distributed RandomForest on Digits Dataset
- Distributed RandomForest on Leukemia Gene Data

Task 2 — Hybrid Big Data Cluster

- Docker Swarm on the same 3 VMs
- Apache Spark (bitnami/spark image)
- PySpark MLlib pipeline
- Distributed RandomForest via Spark
- Spark Web UI monitoring on port 8080

Cluster Architecture

MASTER	WORKER1	WORKER2
192.168.56.10 5 GB 5 cores MPI Rank 0 Swarm Manager	192.168.56.11 4 GB 4 cores MPI Rank 1 Swarm Worker	192.168.56.12 4 GB 3 cores MPI Rank 2 Swarm Worker

Host-Only Network · 192.168.56.0/24 · Passwordless SSH across all 6 node pairs

⇔ MPI / SSH

Task 1 — Distributed ML with MPI

1

Load Dataset

Rank 0 loads Digits / Gene Expression data

2

Broadcast

`comm.bcast()` sends full data to all 6 ranks

3

Shard

Each rank slices its own chunk of samples

4

Train RF

RandomForest (100 trees) trained locally per rank

5

Gather

`comm.gather()` collects accuracy scores at Rank 0

6

Average

Rank 0 prints final average score

```
mpirun -np 6 --hostfile hostfile python3 distributed_mnist.py
```


Task 1 — Results

Digits Dataset

Classifier:	RandomForestClassifier
MPI Processes:	6 (2 per node)
Samples/Rank:	~299
Features:	64 (8×8 pixels)
Classes:	10 (digits 0–9)
Output metric:	Average local accuracy

Leukemia Gene Expression

Classifier:	RandomForestClassifier
MPI Processes:	6 (2 per node)
Samples/Rank:	~33
Features:	500 gene expressions
Classes:	2 (AML vs ALL)
Output metric:	Average local accuracy

Task 2 — Hybrid Cluster: Docker Swarm + Spark



Docker Swarm Init

`docker swarm init on master (192.168.56.10)`



Workers Join

`worker1 & worker2 join via swarm token`



Deploy Stack

`docker stack deploy -c spark-stack.yml spark`



Submit Job

`spark-submit --master spark://master:7077`



Monitor

Spark Web UI → <http://192.168.56.10:8080>

Docker Swarm Services

spark-master

Port: 8080 / 7077

spark-worker-1

Port: 8081

spark-worker-2

Port: 8082

Troubleshooting — Real-World Challenges

● Error 1: Illegal Instruction (Core Dump)

✓ Fix: Set `OPENBLAS_CORETYPE=generic`, reinstall numpy

● Error 2: Externally Managed Environment

✓ Fix: Create venv: `python3 -m venv ~/venvs/hpc_ml`

● Error 3: python3-venv Not Installed

✓ Fix: `sudo apt install python3-venv python3-full`

● Error 4: setuptools.build_meta Missing

✓ Fix: `pip install --upgrade pip setuptools wheel`

● Error 5: mpirun Missing Help Files

✓ Fix: Recompile OpenMPI 4.1.6 from source

● Error 6: prun: proc-aborted (AVX512)

✓ Fix: Compile with `CFLAGS=-mno-avx512f`

● Error 7: undefined symbol: ompi_mpi_errors_abort

✓ Fix: `MPICC=.../mpicc pip install mpi4py`; update PATH

Key Takeaways

HPC vs Big Data

MPI suits tightly-coupled, low-latency tasks. Spark excels at fault-tolerant, large-scale data processing — both complement each other.

Broadcast-Gather Pattern

MPI's bcast/gather enables embarrassingly parallel training: each rank trains independently, then scores are aggregated.

Docker Swarm Simplicity

Swarm provides lightweight orchestration with minimal config. A single YAML file deploys a full Spark cluster across 3 nodes.

Real Environment Challenges

CPU architecture mismatches, PEP 668 restrictions, and broken system packages require deep OS-level troubleshooting skills.

Thank You